

Analyzing Thematic Alignment in Scientific Journals:

A Case Study on the Journal of Machine Learning Research

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Abstract

We present a pipeline for quantifying thematic alignment between papers published in a scientific journal and its Aims & Scope statement. Applied to the Journal of Machine Learning Research (JMLR), we collect 345 abstracts (2015–2024) via the Semantic Scholar API, encode them with Sentence-BERT [1], and compute cosine similarity against the Aims & Scope embedding. BERTopic [2] identifies 9 latent thematic clusters. Mean alignment score: 0.377 ($\sigma = 0.068$); outlier rate: 8.1% (28 papers); thematic drift: $\Delta = -0.001$ over the decade.

Keywords: thematic alignment, sentence embeddings, BERTopic, cosine similarity, JMLR, editorial drift

1 Introduction

Journals communicate their editorial mission through an Aims & Scope statement. Whether published articles actually reflect this mission is rarely measured quantitatively. Dense sentence embeddings [1] enable large-scale semantic similarity comparisons; combined with topic modeling [2], they allow systematic analysis of thematic coherence over time. Related bibliometric work has explored journal scope through citation networks [4] and embedding-based corpus mapping [3].

We apply a four-stage pipeline to JMLR (2015–2024): data collection, Sentence-BERT embedding, cosine similarity scoring, and BERTopic topic modeling. Contributions: (1) per-paper alignment scores against the Aims & Scope; (2) formal outlier detection; (3) temporal drift analysis at corpus and topic level; (4) qualitative validation of extreme cases.

2 Methodology

Research Question. *Do JMLR papers align with the journal’s Aims & Scope, and is there thematic drift between 2015 and 2024?*

Data Collection. 345 abstracts collected via the Semantic Scholar Bulk Search API (<https://api.semanticscholar.org>), filtered by `venue = "Journal of Machine`

Learning Research", year 2015–2024, abstract length ≥ 50 characters. Average abstract length: 1,226 characters. Collection date: June 2026 (2025 papers not yet fully indexed).

Embedding. Abstracts and Aims & Scope encoded with `all-MiniLM-L6-v2` (Sentence-BERT) [1], producing 384-dimensional vectors. Model selected for computational efficiency, open-source availability, and compatibility with HDBSCAN clustering. Embeddings cached as NumPy arrays.

Alignment Score. Cosine similarity between each abstract embedding $\vec{e}_p$ and the Aims & Scope embedding $\vec{e}_{\text{scope}}$:

$$\text{score}(p) = \frac{\vec{e}_p \cdot \vec{e}_{\text{scope}}}{\|\vec{e}_p\| \cdot \|\vec{e}_{\text{scope}}\|} \quad (1)$$

Outlier Detection. Papers with score $< \mu - 1.5\sigma$ classified as outliers:

$$\text{threshold} = 0.377 - 1.5 \times 0.068 = 0.275 \quad (2)$$

Topic Modeling. BERTopic [2] with precomputed embeddings, HDBSCAN (`min_cluster_size=5`, `min_samples=3`), c-TF-IDF representation, target 10 topics.

3 Results

3.1 Dataset

Table 1 shows the year distribution. Volume peaks in 2021 (62 papers); the lower 2024 count reflects incomplete API indexing at collection time.

Table 1: JMLR papers collected per year

Year	Papers	Year	Papers
2015	24	2020	53
2016	21	2021	62
2017	20	2022	39
2018	29	2023	43
2019	32	2024	22

Total: 345

3.2 Alignment Score Distribution

Scores are approximately normally distributed: $\mu = 0.377$, $\sigma = 0.068$, range $[0.123, 0.602]$, median = 0.383. Threshold = 0.275 identifies 28 outliers (8.1%) shown in red in Figure 1.

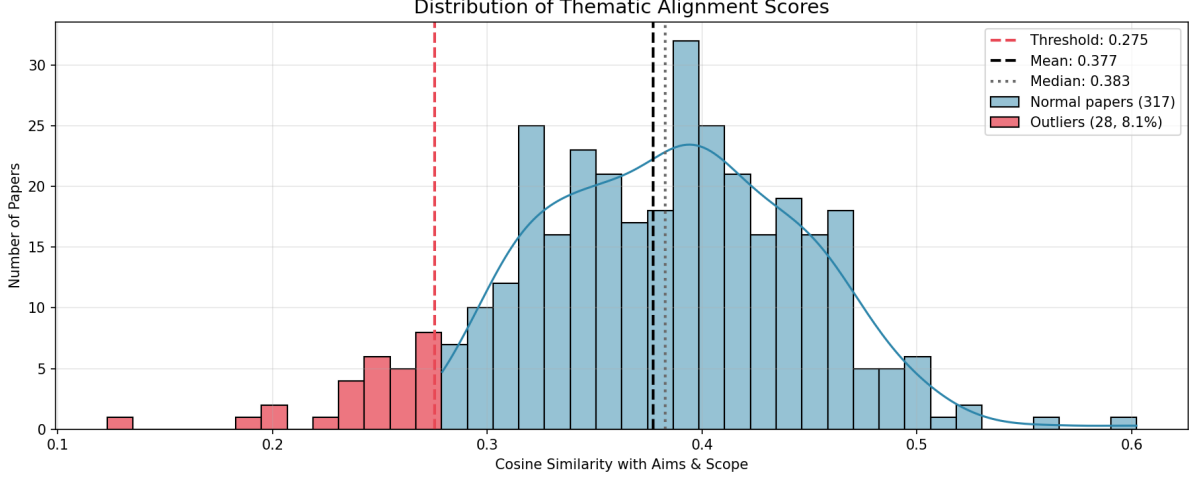


Figure 1: Alignment score distribution. Red: 28 outliers (< 0.275). Dashed: mean (0.377); dotted: median (0.383).

3.3 Outlier Analysis

Table 2 and Figure 2 show the five most and least aligned papers. Top papers are core ML theory contributions: *The Bayesian Learning Rule* (0.602) derives a general Bayesian learning framework; *Distributed Learning with Regularized Least Squares* (0.561) provides theoretical guarantees for distributed optimization.

The 28 outliers fall into three categories:

- **Software toolkits** (7 papers): *Seglearn* (0.225), *depyf* (0.123), *GraSPy* (0.234) — implementation-focused language diverges from theoretical framing.
- **Domain-specific applications** (8 papers): *Extracting PICO Sentences from Clinical Trial Reports* (0.202) applies NLP to a narrow biomedical task.
- **Pure mathematics** (13 papers): *Tensor Perturbation Bound* (0.203), *Multiparameter Persistence Landscapes* (0.238) — specialized vocabulary with minimal ML overlap.

Table 2: Top 5 most and least aligned JMLR papers

Title	Year	Score
<i>Most aligned</i>		
The Bayesian Learning Rule	2021	0.602
Distributed Learning with Regularized Least Squares	2016	0.561
Approximate Newton Methods	2017	0.526
Optimal Rates of Distributed Regression	2020	0.519
An Entropy-Based Model for Hierarchical Learning	2022	0.509
<i>Least aligned</i>		
Seglearn: A Python Package for Learning Sequences	2018	0.225
A Sharp Blockwise Tensor Perturbation Bound	2020	0.203
Extracting PICO Sentences from Clinical Trial Reports	2016	0.202
Consensus-based Optimization on the Sphere II	2020	0.190
depyf: Open the Opaque Box of PyTorch Compiler	2024	0.123

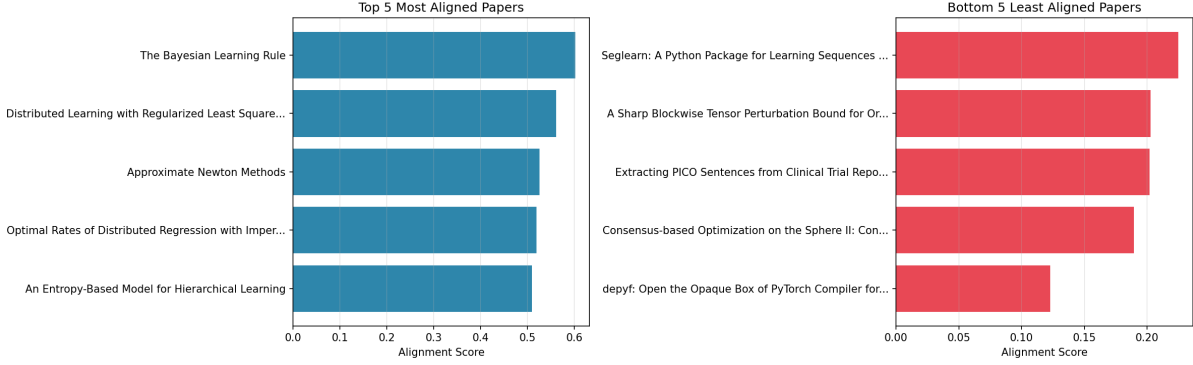


Figure 2: Top 5 most aligned (blue) and bottom 5 least aligned (red).

3.4 Thematic Drift

Figure 3 shows mean and median scores per year. Both are stable ($[0.365, 0.392]$) with no sustained trend. Table 3 confirms: $\Delta = -0.001$ between early and late period, negligible vs. $\sigma = 0.068$.

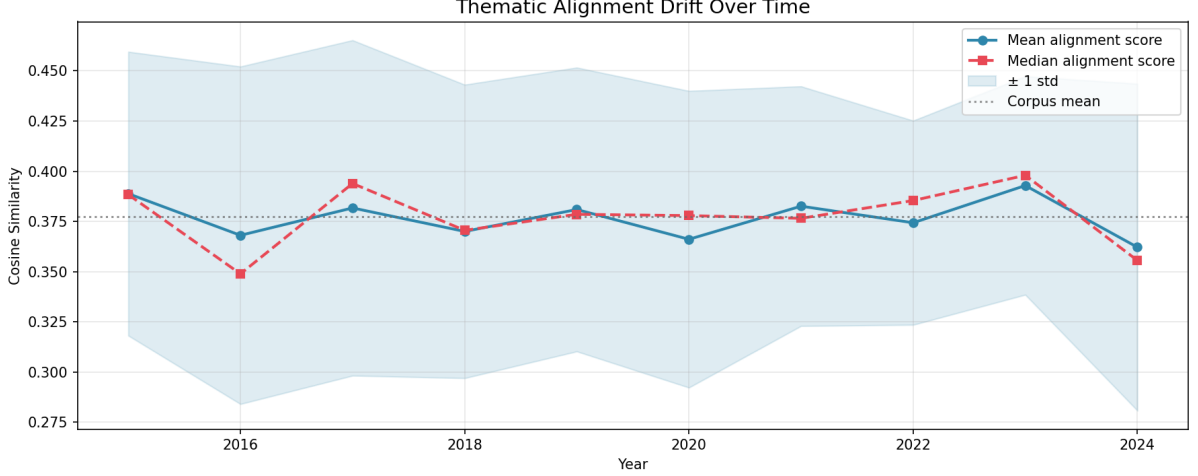


Figure 3: Mean (blue) and median (red dashed) alignment per year, $\pm 1\sigma$ band. No drift detected over 2015–2024.

Table 3: Alignment score by period

Period	Papers	Mean	Std
2015–2019	126	0.378	0.071
2020–2024	219	0.377	0.066
Δ	—	−0.001	—

3.5 Topic Modeling

BERTopic identified 9 clusters (Table 4). Topic 0 (Optimization, 107 papers) is largest and most aligned (0.401); Topic 4 (Graph Learning) scores lowest (0.312). Software & Toolkits (Topic 3, 0.358) also below average, consistent with outlier analysis.

Table 4: BERTopic clusters with alignment scores

ID	Name	Papers	Score
0	Optimization & Convergence Theory	107	0.401
1	Neural Networks & Probabilistic Models	66	0.375
2	Causal & Statistical Learning	28	0.362
3	Software & Toolkits	21	0.358
4	Graph Learning	20	0.312
5	Bandits & AutoML	15	0.380
6	Meta-Learning & AutoML	10	0.398
7	Clinical & Treatment Learning	9	0.358
8	Linear Algebra & Matrix Theory	8	0.348

Figure 4 shows topic frequency over time: Topic 0 peaks in 2021; Topic 1 (Neural Networks) rises from 2021 onward. Figure 5 shows per-topic alignment is stable across

all years, with Topic 0 consistently above and Topics 3–4 consistently below the corpus mean.

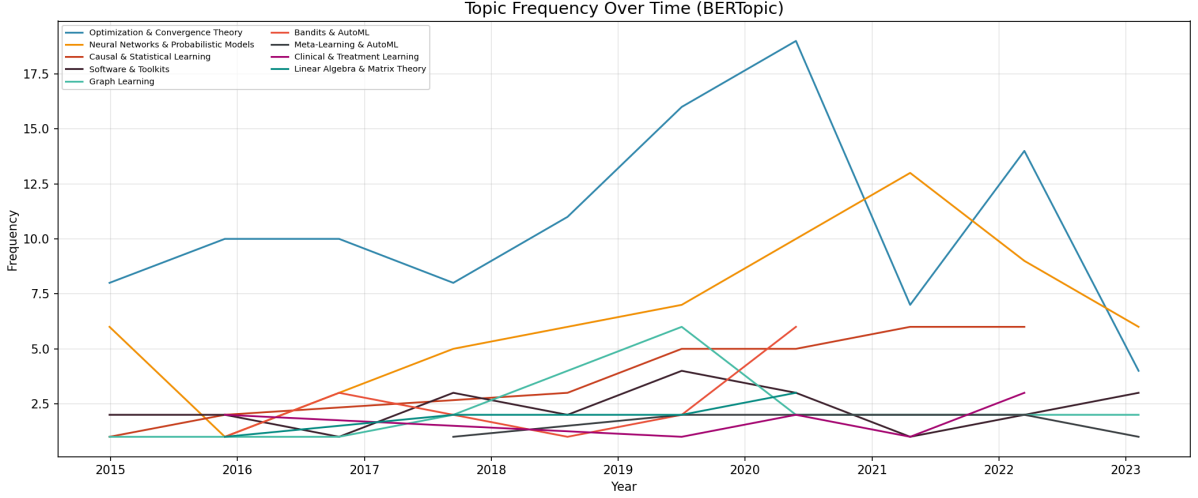


Figure 4: Topic frequency over time. Topic 0 dominates; Topic 1 grows from 2021.

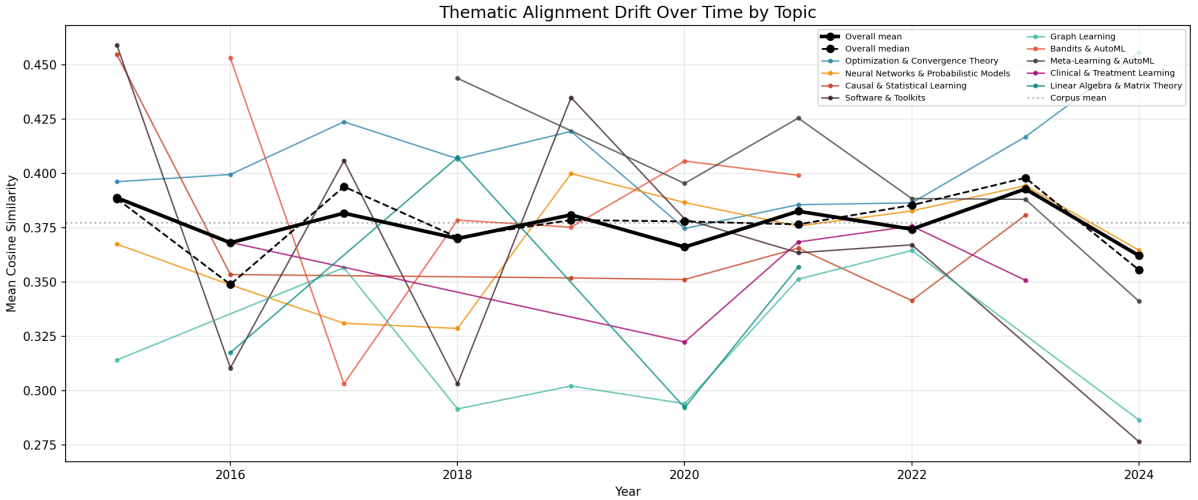


Figure 5: Mean alignment per year by topic. Topic 0 above corpus mean throughout; Topics 3 and 4 persistently below.

4 Concluding Remarks

The pipeline yields three main findings. (1) JMLR shows moderate thematic coherence ($\mu = 0.377$, $\sigma = 0.068$) with 8.1% outliers, concentrated in software toolkits, domain-specific applications, and pure mathematics — predictable edge cases for a broad ML journal. (2) No editorial drift is detected ($\Delta = -0.001$ over ten years). (3) Optimization-focused papers (Topic 0) are most aligned; Graph Learning (Topic 4) and Software (Topic 3) least aligned, stably so across the decade.

Limitations. Cosine similarity measures surface semantic overlap, not conceptual depth. all-MiniLM-L6-v2 was trained on general English, not ML-specific text. Coverage is limited to Semantic Scholar-indexed abstracts (345 of an estimated 500+ papers). The Aims & Scope is a static vector, ignoring possible policy revisions.

Future Work. SciBERT or SPECTER embeddings for domain-specific sensitivity; multi-journal comparison; full-text analysis beyond abstracts.

AI Usage Disclaimer. Claude (Anthropic) was used as a debugging assistant during implementation and also to help with explanations.

References

- [1] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proc. EMNLP-IJCNLP 2019*, pp. 3982–3992.
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